

FloodSense — Feature Specifications & Complete System Flow

FLOODSENSE

AR Flood Susceptibility Mobile Application

Feature Specifications & Complete System Flow

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1. Document Overview

This document details all features of the FloodSense mobile application, including the newly proposed features added in response to the panel's feedback during the thesis defense. It covers the complete user flow from app launch to every available interaction, the purpose of each feature, implementation notes, and feasibility rationale.

FloodSense was originally developed as a pre-event flood awareness tool that combines machine learning-based susceptibility classification with augmented reality visualization. Following the panel's challenge — 'What happens next after users become aware?' — three additional features were designed to transition the system from passive awareness to active, actionable guidance.

2. Complete System Flow

The following is the complete step-by-step flow a user experiences from opening the app for the first time to accessing every available feature in FloodSense.

2.1 First-Time Launch Flow

STEP 1 — App Launch

User opens FloodSense on their Android device.

STEP 2 — Onboarding & Disclaimer Screen

First-time users are shown the onboarding screen explaining the app's purpose, scope, and limitations. A disclaimer states that susceptibility levels are modeled pre-event estimates, not real-time flood guarantees. Flat-surface AR instructions are also shown here. User taps 'I Understand' to proceed.

STEP 3 — Login / Registration Screen

User either logs in with existing username and password or registers a new account. Firebase Authentication handles credential management using the surrogate email workaround. New users provide a username, password, and their home barangay.

STEP 4 — Main Dashboard / Home Screen

After login, the user lands on the main screen with access to all core modules: Rainfall Input, AR Map, Results Summary, Account & Preferences, and the newly added action features.

2.2 Core Feature Flow (Existing Features)

STEP 5 — PAGASA-Aligned Rainfall Input

User selects rainfall intensity (Light, Moderate, Heavy, Intense, Torrential) and duration (1, 3, 6, 12, 24 hours) from plain-language dropdown menus. The app internally maps these to mm/hr values used by the ML model. Saved preferences from a previous session are pre-filled automatically.

STEP 6 — ML Susceptibility Classification

The app retrieves the pre-computed susceptibility lookup table from Firebase Firestore based on the selected rainfall intensity and duration combination. No live model inference occurs at runtime. Each barangay is assigned a susceptibility level: Low, Moderate, High, or Very High.

STEP 7 — AR Visualization

User taps 'View AR Map'. The camera activates and Vuforia's ground-plane engine scans for a flat surface. The surface detection indicator shows red (invalid) or green (valid). Once a valid surface is confirmed, the color-coded barangay susceptibility map is anchored to the surface as a ground-plane overlay. User can pinch-to-zoom to examine individual barangay labels. A color legend and disclaimer remain accessible during AR view.

STEP 8 — Results Summary Screen

A scrollable list of all Bacoor City barangays with their color-coded susceptibility badges is displayed. This provides an at-a-glance overview without requiring the AR interface.

2.3 New Feature Flow (Post-Awareness Action Features)

After viewing the AR map or the Results Summary, the following new action buttons are presented to the user, transitioning the app from passive awareness to active guidance.

STEP 9 — Check My Area (Safety Status Check)

User taps 'Check My Area'. A map of Bacoor City is displayed with a draggable pin marker. The user manually drags the pin to their preferred location. No GPS permission is required. Once the pin is placed, the app cross-references the pin's coordinates against a GeoJSON boundary file of Bacoor City's barangays to identify the corresponding barangay. A dedicated results screen is then displayed for that specific pinned location, showing: the identified barangay name, the active rainfall scenario (intensity and duration), the corresponding susceptibility level with its color

indicator, and a plain-language safety status summary. The result is tied to the rainfall scenario currently selected by the user — if the user returns and changes the rainfall scenario, the result updates automatically for the same pinned location without requiring the user to re-pin.

STEP 10 — What Should I Do? (Action Checklist)

Based on the susceptibility level returned for the user's barangay, a tailored action checklist is shown. For Low: monitor and prepare. For Moderate: secure valuables and identify evacuation routes. For High/Very High: proceed to evacuation if advised, avoid flood-prone roads, and contact relatives.

STEP 11 — Find Nearest Evacuation Center

Triggered automatically when susceptibility is Moderate, High, or Very High. A ranked list of BDRRMO-designated evacuation centers is displayed, sorted by distance from the user's GPS location using the Haversine formula. Each entry shows the center name, barangay, and estimated distance. User taps 'Get Directions' and is deep-linked to Google Maps for turn-by-turn navigation.

2.4 Supporting Features

Account & Preferences — User Account Management

Users can view and update their registered barangay and saved rainfall preferences. Only one set of rainfall preferences is saved per account. Login is username and password only; social login is not supported.

Onboarding (Repeat Access) — Instructions & Disclaimer Revisit

Users can revisit the onboarding instructions and disclaimer at any time from the main screen via 'View Instructions and Parameters'.

3. Feature Specifications

Feature 1 — AR Flood Susceptibility Map (Existing)

The core visualization feature of FloodSense. Renders a color-coded, barangay-level flood susceptibility map anchored to any detected flat surface through the device camera using Vuforia ground-plane markerless tracking.

Aspect	Details
Status	Existing core feature
Technology	Unity + Vuforia SDK (Ground Plane)
Input	Pre-computed susceptibility lookup table retrieved from Firebase Firestore based on user-selected rainfall parameters

Output	Color-coded AR map overlay: Low (green), Moderate (yellow), High (orange), Very High (red)
Surface Detection	Green indicator = valid flat surface detected; Red = invalid. Map is locked until surface is confirmed green.
Legend	Visible within AR view. Full metadata accessible via in-app info panel.
Disclaimer	Accessible from AR view at all times. States susceptibility is a modeled estimate, not a real-time flood guarantee.
Interaction	Pinch-to-zoom to examine individual barangay labels and susceptibility colors.
GPS Used	No. AR overlay is anchored to a flat surface, not GPS-linked to outdoor position.

Feature 2 — PAGASA-Aligned Rainfall Input (Existing)

Users specify a rainfall scenario through plain-language dropdown menus aligned with PAGASA's public advisory classification. The app handles all internal mm/hr conversion transparently.

Aspect	Details
Status	Existing core feature
Intensity Options	Light (<2.5 mm/hr), Moderate (2.5-7.5 mm/hr), Heavy (7.5-15 mm/hr), Intense (15-30 mm/hr), Torrential (>30 mm/hr)
Duration Options	1 hour, 3 hours, 6 hours, 12 hours, 24 hours
Design Rationale	Plain-language labels remove the need for technical meteorological knowledge. Validated by BDRRMO key informant.
Saved Preferences	One set of rainfall preferences saved per user account via Firebase Firestore.

Feature 3 — ML Susceptibility Classification (Existing)

A supervised machine learning model trained offline on environmental datasets classifies flood susceptibility per barangay. Results are pre-computed and stored as static JSON lookup tables in Firestore — no live model inference occurs at runtime.

Aspect	Details
Status	Existing core feature
Algorithm	Random Forest or XGBoost (scikit-learn / Python)
Input Features	Elevation, historical flood records (BDRRMO), rainfall intensity and duration (PAGASA-aligned)
Output Classes	Low, Moderate, High, Very High — one classification per barangay per rainfall scenario
Runtime Behavior	App retrieves pre-computed JSON from Firestore. No ML server required at runtime.
Data Sources	BDRRMO historical flood records, CPDCO elevation and land use maps, PAGASA rainfall classification tables
Known Limitation	Model trained on static environmental conditions. Does not account for real-time drainage blockages, soil saturation, or river water levels.

Feature 4 — Results Summary & Decision Support (Existing)

A scrollable list of all Bacoar City barangays with color-coded susceptibility badges, populated from the pre-computed Firestore lookup table. Provides an at-a-glance overview without requiring the AR interface.

Aspect	Details
Status	Existing core feature
Technology	Unity uGUI ScrollView / List
Data Source	Pre-computed Firestore JSON document keyed by selected rainfall intensity and duration
Current Limitation	No sorting, filtering, or search functionality. All barangays displayed in a flat list.

Feature 5 — Check My Area: Location-Based Safety Status (NEW)

Addresses the panel's requirement to inform users whether their current location is safe under the selected rainfall scenario. After viewing the susceptibility map, the user taps 'Check My Area' and is presented with a map of Bacoar City containing a draggable pin marker. The user manually positions the pin on their preferred location

— no GPS permission is required. Once the pin is placed, a dedicated results screen is shown for that specific pinned location, reflecting the rainfall scenario currently selected by the user. The results display the identified barangay, the active rainfall intensity and duration, the susceptibility level with its color indicator, and a plain-language safety status summary. The result is scenario-aware: if the user changes the rainfall scenario, the output updates automatically for the same pinned location without requiring re-pinning.

Aspect	Details
Status	NEW — proposed in response to panel feedback
Trigger	User taps 'Check My Area' button after viewing the AR map or Results Summary
Technology	Interactive map with draggable pin marker + GeoJSON barangay boundary file + existing Firestore lookup table
User Interaction	A map of Bacoor City is displayed with a draggable pin. The user drags and drops the pin onto their location. No GPS permission is requested or required.
Location Detection	The pin coordinates upon release are cross-referenced against the GeoJSON barangay boundary file to identify the barangay. All processing is done locally on the device.
Rainfall Scenario Linkage	The result is dynamically tied to the rainfall scenario currently selected by the user (intensity and duration). Changing the scenario refreshes the result for the same pinned location automatically.
Result Screen Content	Identified barangay name, active rainfall scenario (intensity + duration), susceptibility level with color indicator (Low/Moderate/High/Very High), and a plain-language safety status summary specific to that barangay and scenario.
Data Privacy	Compliant with RA 10173. No coordinates are transmitted or stored — pin placement and result computation are performed locally within the session only.
Example Output (Low)	Pinned Location: Brgy. Zapote Scenario: Heavy rainfall, 3 hours Status: LOW susceptibility. Stay alert and monitor updates.

Example Output (High)	Pinned Location: Brgy. Talaba Scenario: Heavy rainfall, 3 hours Status: HIGH susceptibility. Consider taking precautionary action.
Feasibility	High. The existing Firestore lookup table and GeoJSON boundary file already support this flow. The scenario-linked result requires no additional data infrastructure — only a result screen that reads from the already-queried lookup table entry for the resolved barangay and selected scenario.

Feature 6 — What Should I Do? Action Checklist (NEW)

Addresses the panel's concern that awareness alone is insufficient and that the system must educate and guide the user's decision-making. A static, level-specific checklist appears after the safety status card, providing concrete pre-event actions the user should take.

Aspect	Details
Status	NEW — proposed in response to panel feedback
Trigger	Automatically displayed after the Safety Status card (Feature 5) is shown
Technology	Static content stored in app or Firestore, mapped to susceptibility levels. No additional logic or API required. A conversational AI assistant capable of answering user queries in the context of their susceptibility result is identified as a viable future enhancement, subject to API cost evaluation and institutional approval.
Implementation Complexity	Very low. Essentially a content screen with four preset text sets mapped to Low, Moderate, High, Very High.

Checklist Content by Susceptibility Level

LOW Susceptibility:

- Monitor weather updates from PAGASA and BDRRMO.
- Prepare your go-bag with essentials (documents, medicine, water, clothes).
- Check drainage around your property.

MODERATE Susceptibility:

- Secure important documents and valuables in waterproof storage.
- Identify your nearest evacuation center.
- Prepare your go-bag and keep it accessible.
- Inform family members of your location and plan.

HIGH Susceptibility:

- Proceed to an evacuation center if advised by BDRRMO or barangay officials.
- Avoid flood-prone roads and low-lying areas.
- Bring your go-bag and essential documents.
- Alert relatives and neighbors about your situation.
- Follow all official BDRRMO and PAGASA advisories.

VERY HIGH Susceptibility:

- Evacuate immediately if not already done.
- Do NOT attempt to cross flooded roads or waterways.
- Bring go-bag, documents, medicine, and sufficient food and water.
- Contact relatives and inform them of your evacuation destination.
- Follow BDRRMO and PAGASA emergency directives without delay.

Feature 7 — Find Nearest Evacuation Center (NEW)

Addresses the panel's requirement to automatically direct users to the nearest evacuation centers. Triggered when the user's barangay is classified as Moderate, High, or Very High susceptibility.

Aspect		Details
Status		NEW — proposed in response to panel feedback
Trigger		Automatically offered after Safety Status card shows Moderate, High, or Very High result
Technology		Static evacuation center JSON stored in Firestore + Haversine formula for distance calculation + Google Maps URI deep-link
Evacuation Data	Center	Sourced from BDRRMO as part of existing data-sharing collaboration. One-time data entry of center names, barangays, and GPS coordinates into Firestore.

Distance Calculation	Haversine formula computed on the device using the pin's dropped coordinates (from Feature 5) and each center's stored coordinates. No routing API or paid service required.
Display	Ranked list of nearest evacuation centers showing: center name, barangay, estimated straight-line distance from user.
Directions	Each center entry has a 'Get Directions' button that deep-links into Google Maps (via geo: URI or maps.google.com URL). All routing is handled by Google Maps — no custom routing engine built.
Location Source	Coordinates from the pin placement in Feature 5 (Check My Area). Not stored or transmitted.
Feasibility	High. No paid API required. Google Maps deep-link is a standard Android feature. Evacuation center list is a one-time static dataset obtainable from BDRRMO.

4. Consolidated App Flow Summary

The following summarizes the complete end-to-end flow of FloodSense from first launch through all available features.

#	Step / Feature	What Happens
1	App Launch	User opens FloodSense on Android device
2	Onboarding & Disclaimer	First-time users read app purpose, scope, flat-surface instructions, and disclaimer. Tap 'I Understand.'
3	Login / Register	Existing users log in. New users register with username, password, and home barangay.
4	Main Dashboard	Home screen with access to all modules.
5	Rainfall Input	User selects PAGASA-aligned intensity and duration from dropdowns.
6	ML Classification	App retrieves pre-computed susceptibility lookup from Firestore for selected rainfall scenario.
7	AR Map View	Camera activates. Ground-plane detected. Color-coded susceptibility map rendered on flat surface.

8	Results Summary	Scrollable barangay list with color-coded susceptibility badges.
9	Check My Area (NEW)	Map with draggable pin displayed. User drags pin to their location. Barangay resolved from pin coordinates. Results screen shown with barangay name, active rainfall scenario, susceptibility level, and plain-language safety summary.
10	Action Checklist (NEW)	Level-specific 'What Should I Do?' checklist shown based on susceptibility result.
11	Nearest Evacuation Center (NEW)	Ranked list of BDRRMO evacuation centers sorted by distance. 'Get Directions' deep-links to Google Maps.
12	Account & Preferences	User manages registered barangay and saved rainfall preferences.
13	Revisit Onboarding	User can re-read instructions and disclaimer from main screen at any time.

5. New Features Feasibility Summary

Feature	Complexity	Est. Dev Time	Key Dependency
Check My Area (Safety Status)	Low-Medium	1 week	Interactive map with draggable pin + GeoJSON barangay boundary file + existing Firestore lookup
Action Checklist	Very Low	2–3 days	Static content only, no new API or data source required
Find Nearest Evacuation Center	Low-Medium	1 week	Evacuation center list from BDRRMO + Google Maps deep-link

All three new features operate within the existing Firebase free tier and require no additional paid APIs or services. Combined estimated development time for all new features is approximately 2.5–3 weeks, which is well within an 8-month development timeline.

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